THE DATA REFINERY — 21-DAY SPRINT

Author: Prasad S Hiremath | **Date:** October 12, 2026 | **Day 4:** SQL Window Functions & Joins

1. Core Concepts: Window Functions vs. GROUP BY

What is a Window Function? Unlike GROUP BY (which collapses rows into summary groups), a Window Function performs calculations across related table rows while **preserving individual row granularity** in the output.

Syntax Structure:

```
FUNCTION_NAME() OVER (PARTITION BY column1 ORDER BY column2)
```

Feature	GROUP BY	Window Function (OVER)
Row Count	Reduced (collapsed into groups)	Preserved (matches input row count)
Granularity	Group-level	Row-level + Group metrics
Primary Use	Totals, averages, counts per group	Ranking, deduplication, LAG/LEAD, running totals

2. SQL Coding Challenge (Joins & Aggregations)

Scenario: Write an INNER JOIN query to list each category name along with the total stock quantity of products in that category.

```
SELECT c.category_name, SUM(p.stock_qty) AS total_stock FROM categories c INNER JOIN products p ON c.category_id = p.category_id GROUP BY c.category_name;
```

3. Interview MCQs & Detailed Solutions (5 Core + 2 Bonus)

Q1: Deduplication: Which window function assigns a unique sequence number to duplicate records for filtering (`ROW_NUMBER = 1`)?

• **Answer:** C) ROW_NUMBER()
• **Reasoning:** ROW_NUMBER() assigns a strictly unique sequential integer to each row within a partition regardless of ties.

Q2: Ranking Ties: If two employees tie for top salary, what do RANK() and DENSE_RANK() produce for the next rank?

• **Answer:** B) RANK() produces 3; DENSE_RANK() produces 2
• **Reasoning:** RANK() skips rank numbers after ties (1, 1, 3), whereas DENSE_RANK() does not skip numbers (1, 1, 2).

Q3: Time-Series Shift: What does LAG(closing_price, 1) OVER (ORDER BY trade_date) return for the very first row?

• **Answer:** C) NULL
• **Reasoning:** Because no preceding row exists before the first record, LAG() safely defaults to returning NULL.

Q4: Row Preservation: True or False: Window functions reduce the total number of rows returned in the final result set.

• **Answer:** B) False
• **Reasoning:** Window functions output analytic results alongside every individual source row, keeping row counts identical.

Q5: Future Lookup: To calculate the time gap between a user's current login and their NEXT login, which function to use?

• **Answer:** B) LEAD()
• **Reasoning:** LEAD() accesses data from a subsequent (future) row, whereas LAG() accesses preceding (past) rows.

Q6 (Bonus 1): Default Framing: What is the default window frame when ORDER BY is specified without an explicit ROWS/RANGE clause?

• **Answer:** B) ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
• **Reasoning:** Standard ANSI SQL defaults to a cumulative frame from the start of the partition up to the current row.

Q7 (Bonus 2): Running Totals: Which expression correctly computes a cumulative running total of daily revenue without collapsing rows?

• **Answer:** A) SUM(revenue) OVER (ORDER BY date)
• **Reasoning:** Using SUM() with an OVER (ORDER BY date) clause creates a running cumulative sum across ordered rows.